Reliable Policy Transfer for Safety-Aware End-to-End Driving with Deep Reinforcement Learning

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https://github.com/

Abstract

End-to-End (E2E) Reinforcement Learning (RL) for autonomous driving still struggles with safety and generalization under distribution shift, as perception-heavy encoders, sparse rewards, and ad hoc uncertainty handling often yield brittle closed-loop behavior. This work introduces a unified Deep RL (DRL) framework addressing key gaps: causal ego-centric state design, dense differentiable rewards, joint uncertainty estimation with entropy gating, and control-level policy transfer. An ego-centric relational graph encodes agent influence via uncertainty-weighted attention over kinematics, lane geometry, and semantics, producing a compact control state. A multi-objective differentiable reward stabilizes optimization by shaping safety, progress, and comfort with an uncertainty term. Aleatoric and epistemic uncertainty-captured through per-edge heteroscedastic variance and a critic ensemble-are aggregated into a calibrated confidence signal that modulates policy entropy for risk-aware exploration. A causal-semantic transfer objective aligns actions, attention, and uncertainty statistics across domains, combined with meta-learned initialization for few-shot adaptation. In closed-loop urban driving across varied towns, traffic, and weather, the framework improves success rate, reduces infractions per kilometer, and achieves higher time-to-conflict with lower lateral deviation and comfort cost compared to strong baselines.

1. Introduction

Autonomous driving has advanced from modular perception-planning pipelines to End-to-End (E2E) systems that couple scene understanding with control. Recent works demonstrate strong open-loop accuracy and scalable architectures [5, 10, 11, 14], yet robustness under distribution shift, dense traffic, and adverse weather remains limited [3]. Benchmarks increasingly emphasize online metrics such as success rate and infractions per kilometer, revealing safety and generalization gaps not captured by open-loop evaluation [15]. Common limitations across E2E stacks include: state encoders compress perception without causal relations or confidence [9], rewards are sparse or discontinuous in safety-critical control [18], uncertainty is inconsistently estimated and rarely regulates exploration [1], and transfer emphasizes perception features rather than control-level reasoning [4, 13, 20].

This paper introduces a unified Deep Reinforcement Learning (DRL) framework with four complementary innovations. First, an ego-centric relational representation makes influence explicit by weighting interactions with distance and uncertainty. Second, a differentiable multi-objective reward provides dense, physically grounded feedback for safety, progress, and comfort, stabilizing training. Third, joint aleatoric-epistemic estimation is coupled to the policy via entropy gating, aligning exploration with confidence [20]. Fourth, a causal-semantic policy transfer objective regularizes action distributions, relational attention, and uncertainty statistics across domains [13], paired with meta-learned initialization for rapid adaptation [21].

The method instantiates an ego-node relational graph that aggregates edge features, lane geometry, and aleatoric variance via uncertainty-weighted attention, yielding a compact control state for the actor-critic. Reward shaping replaces thresholded events with smooth surrogates. Decision-time uncertainty is divided into aleatoric and epistemic components [9], with ensemble-based estimation further stabilizing confidence [1]. A calibrated summary modulates policy entropy, reducing risk under low confidence and restoring exploration when confidence increases. For transfer, a regularizer combines policy Kullback-Leibler (KL) divergence, Maximum Mean Discrepancy (MMD) alignment, and moment matching of uncertainty statistics [13, 20], paired with a Model-Agnostic Meta-Learning (MAML) initialization for rapid few-shot adaptation [21].

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The framework is validated in closed-loop urban driving across diverse towns, traffic densities, and weather regimes representative of realistic evaluation settings [15]. Experiments show consistent gains in success rate and notable reductions in infractions per kilometer compared to strong E2E baselines [11], along with higher minimum time-to-conflict and improved lateral adherence and comfort [4]. Ablations indicate that uncertainty-weighted attention and entropy gating are both essential for robust safety under sensor noise and crowding, while the differentiable reward enhances optimization stability and transfer [26]. Causal-semantic alignment further narrows the generalization gap across domains, enabling reliable few-shot adaptation without sacrificing comfort or efficiency [13, 20].

This framework offers the following key contributions:

- Ego-centric relational state that fuses semantics and geometry into a compact control representation (Sec. 3.3).
- Differentiable reward that optimizes safety, progress, and comfort through uncertainty-aware shaping (Sec. 3.4).
- Uncertainty-transfer mechanism that aligns exploration with confidence and regularizes reasoning across domains, paired with meta-initialization (Secs. 3.5–3.6).

2. Related Work

E2E driving with RL has advanced rapidly, but still struggles with state representation, reward shaping, uncertainty, and policy transfer, as discussed in following subsections.

2.1. Causal State Representation

Early E2E stacks emphasize perception and fusion but leave the control layer unaware of which entities causally influence the ego vehicle [5, 10]. Formulations such as UniAD and ThinkTwice improve task coupling, yet interactions are still encoded through global feature tensors rather than explicit ego-centric relations [11, 14]. Vectorized and Bird's-Eye-View (BEV) representations enhance efficiency and geometry alignment but provide limited interpretability of edge-wise influences tied to safety outcomes [16, 22]. Recent task decoupling reduces perception-planning entanglement but does not expose uncertainty-weighted attention to prioritize interactions by distance and confidence [5, 13].

World models and language-augmented planners enrich scene semantics but emphasize capacity over compact, control-ready causal structure [23–25]. Bias analyses reveal entanglement between geometry and control supervision, complicating safety [12]. Graph-style reasoning appears in prediction and planning, yet a unified state fusing lane geometry, goal progress, ego dynamics, and uncertainty into a single interpretable vector remains uncommon [4, 6]. These inspire an ego-node relational graph that encodes causal and uncertainty-weighted interactions in a compact state to make safety factors explicit for policy learning.

2.2. Multi-Objective Reward Shaping

Reward shaping for E2E driving often relies on sparse or thresholded penalties, causing discontinuities and weak gradients during optimization [8, 18]. Closed-loop studies show that progress-only or event-driven rewards encourage brittle behaviors in dense traffic and intersections [14, 15]. Richer supervision via prediction and planning heads was introduced, yet control objectives remain under-specified for smoothness and comfort [4, 11]. Imitation-augmented methods ease reward design but inherit suboptimal biases and lack risk sensitivity under distribution shift [8, 12].

Differentiable multi-objective formulations improve stability when safety, progress, and smoothness are optimized jointly, though uncertainty is rarely integrated into continuous reward terms [4, 27]. World model and hierarchical approaches enhance long-horizon credit assignment but still require domain-stable shaping to generalize across towns and weather [21, 23, 25]. Large-scale benchmarks emphasize closed-loop success and infractions per kilometer, reinforcing the need for dense, physically meaningful rewards that generalize [15]. These observations inspire differentiable multi-objective shaping that integrates driving objectives to ensure stable gradients and domain-stable control.

2.3. Uncertainty Estimation for Exploration

Uncertainty-aware driving has explored distributional critics and ensembles to capture predictive dispersion, yet most systems treat aleatoric and epistemic components separately rather than jointly at decision time [1, 9]. Continual learning frameworks incorporate confidence but rarely propagate uncertainty through the scene representation used by the controller [5, 19]. Without uncertainty-weighted attention, policies risk overreacting to noisy detections or underweighting high-risk actors [10, 16]. Entropy in RL is scheduled globally rather than gated by real-time confidence, weakening the link between risk and exploration [9, 17].

Safety gains have been reported when uncertainty modulates prediction and planning heads, though consistent connections to control entropy remain limited in E2E stacks [4, 24]. Bias analyses of imitation policies further highlight the need to expose confidence pathways to the decision module [12]. Joint estimation of aleatoric and epistemic uncertainty, propagated along relational edges, can inform both attention and adaptive entropy for cautious exploration when confidence is low [1, 9]. These gaps stimulate a unified mechanism that estimates both uncertainty types, routes them through the ego-centric graph, and gates policy entropy to align risk tolerance with confidence.

2.4. Semantic Alignment in Policy Transfer

Policy transfer often adapts perception features while leaving causal cues and uncertainty statistics misaligned across domains [13, 20]. Evaluations show overfitting to training

towns or weather, with limited attention consistency in new scenes [11, 15]. Language-based adaptation and decoupling improve portability but do not explicitly align relational attention patterns tied to safety-critical interactions [13, 20]. Meta-learning accelerates adaptation yet is rarely combined with uncertainty statistics to guide shot updates [21, 23].

Causal consistency can be promoted by matching policy distributions and attention statistics across domains, rather than pixel or BEV features [5, 16]. Uncertainty statistics are likewise transferable signals that should be regularized to stabilize exploration and comfort after domain shift [9, 19]. Robotics studies in navigation indicate that semantics and uncertainty-informed transfer reduce drift and improve safety across simulators, motivating analogous designs in driving [2, 7]. These insights guide a causal-semantic transfer to achieve few-shot policy adaptation.

3. Unified End-to-End Policy Transfer

This section introduces a unified DRL framework for autonomous driving. Sec. 3.1 formalizes the Markov Decision Process (MDP). Sec. 3.2 defines the closed-loop driving problem and constraints. Sec. 3.3 presents an ego-node relational state. Sec. 3.4 develops a multi-objective differentiable reward. Sec. 3.5 derives joint uncertainty estimation with entropy gating. Sec. 3.6 presents a causal-semantic transfer objective with meta-learned initialization.

3.1. Preliminaries

This work models closed-loop driving as an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, r, P, s_0, \gamma)$ with time-discretized control. The state space $\mathcal{S}$ contains ego-centric scene descriptions at time t, $s_t \in \mathcal{S}$. The action space $\mathcal{A}$ comprises continuous control commands $a_t = [a_t^{\text{thr}}, a_t^{\text{brk}}, a_t^{\text{str}}] \in \mathcal{A}$ for throttle, brake, and steering. The transition dynamics $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ define $s_{t+1} \sim P(\cdot \mid s_t, a_t)$, capturing vehicle kinematics and interactions with other agents. The reward function $r : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ evaluates safety, progress, smoothness, and uncertainty at each transition (s_t, a_t, s_{t+1}). The process starts at $s_0 \in \mathcal{S}$ and uses a discount factor $\gamma \in [0, 1)$.

A stochastic policy $\pi_\theta : \mathcal{S} \to \Delta(\mathcal{A})$ produces $a_t \sim \pi_\theta(\cdot \mid s_t)$ and is trained to maximize the discounted return

$$\max_\theta \; J(\theta) = \mathbb{E}_{\pi_\theta, P}\left[\sum_{t=0}^{\infty} \gamma^t \, r(s_t, a_t, s_{t+1})\right], \qquad (1)$$

where the expectation is over $a_t \sim \pi_\theta(\cdot|s_t)$ and $s_{t+1} \sim P(\cdot|s_t, a_t)$. Soft value functions V^π and Q^π are used in entropy-regularized actor-critic training, with automatic temperature adjustment for exploration-exploitation balance and target networks for stable critic updates.

For each entity i, relative kinematics are Δp_i (ego-frame position) and Δv_i (relative velocity), semantic class is c_i (vehicle, pedestrian, traffic light), and local lane features are κ_i (curvature, heading offset). Per-entity aleatoric variance is σ_i^2. Decision-time uncertainty is decomposed into aleatoric and epistemic components, σ_{ale}^2 and σ_{epi}^2, aggregated into a normalized summary $\bar{\sigma} \in [0, 1]$ that gates policy entropy. Lane adherence uses lateral deviation d_L to the centerline and a context-dependent membership $\mu_A \in [0, 1]$ that tightens admissible deviation in fragile regimes. Sec. 3.2 formalizes route tracking, constraints, and the closed-loop objective that grounds the subsequent design.

3.2. Problem Formulation

The closed-loop task is to follow a route $\mathcal{R} = \{r_j\}_{j=1}^N$ along lane centerlines while interacting with agents and obeying traffic rules. At each step t, perception yields o_t, and a decision state $s_t \in \mathcal{S}$ is constructed for control as defined in Sec. 3.1. The policy samples $a_t = [a_t^{\text{thr}}, a_t^{\text{brk}}, a_t^{\text{str}}]$, the environment transitions to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$, and the reward $r(s_t, a_t, s_{t+1})$ provides dense feedback. Control ensures adherence, avoids collisions, respects signals, and maintains comfort under partial observability.

Lane adherence is enforced on the active segment (r_j, r_{j+1}) using the lateral point-to-line distance

$$d_L(p_t;\, r_j, r_{j+1}) \;=\; \frac{\|(r_{j+1} - r_j) \times (p_t - r_j)\|_2}{\|r_{j+1} - r_j\|_2}, \qquad (2)$$

where p_t denotes the ego position at time t in the map frame. Admissible deviation is tightened in fragile contexts through a membership-controlled corridor,

$$d_L \;\le\; \epsilon(\mu_A), \quad \epsilon(\mu_A) \;=\; \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})\, \mu_A, \qquad (3)$$

with $\mu_A \in [0, 1]$ computed from contextual factors such as traffic density, reduced visibility, or sharp curvature.

Progress is measured along the route tangent. Let s_t^{arc} denote the arc-length coordinate of p_t on $\mathcal{R}$, and let $\Delta s_t = s_{t+1}^{\text{arc}} - s_t^{\text{arc}}$ represent the step progress. Safety requires maintaining sufficient separation from the nearest entity $i^\star$,

$$\text{dist}(p_t,\, p_t^{(i^\star)}) \;\ge\; d_{\min}, \qquad (4)$$

and a minimum time-to-collision threshold, expressed as $\tau_t = \frac{\|\Delta p_{i^\star}\|_2}{\max\left(\epsilon_v, (-\Delta p_{i^\star} \Delta v_{i^\star})_+\right)} \ge \tau_{\min}$, where $(x)_+ = \max(x, 0)$ and $\epsilon_v > 0$ prevents degeneracy at low speeds. Signal obedience and stopping-line compliance are incorporated as smooth reward penalties to avoid discontinuities.

Comfort is enforced by limiting jerk and steering rate. Let $j_t = \|a_t^{\text{veh}} - a_{t-1}^{\text{veh}}\|_2 / \Delta t$ be longitudinal jerk from ego acceleration a_t^{veh}, and $\dot{\delta}_t = (a_t^{\text{str}} - a_{t-1}^{\text{str}})/\Delta t$ the steering rate. Comfort bounds require $j_t \le j_{\max}$ and $|\dot{\delta}_t| \le \dot{\delta}_{\max}$, implemented as differentiable shaping terms. Action bounds and actuator limits are enforced by squashing.

The objective is to learn a policy that maximizes closed-loop performance. A Lagrangian relaxation combines dense reward r_{d} with smooth penalties $\phi_{\{\cdot\}}$. Let

$\boldsymbol{\lambda} = [\lambda_L, \lambda_S, \lambda_C]^\top \in \mathbb{R}^3_{\geq 0}$ denote nonnegative multipliers for lane adherence, safety, and comfort, and $\boldsymbol{\phi}_t = \big[\phi_L(d_L, \mu_A),\ \phi_S(\text{dist}, \tau_t),\ \phi_C(j_t, \dot{\delta}_t)\big]^\top$ stack penalties at time t, where ϕ_L penalizes lateral deviation d_L with membership μ_A, ϕ_S penalizes safety violations via distance dist and time-to-conflict τ_t, and ϕ_C penalizes comfort violations through jerk j_t and steering rate $\dot{\delta}_t$. The objective is

$$\max_{\theta}\ \mathbb{E}\left[\sum_{t\geq 0}\gamma^t\Big(r_{\text{d}}(s_t, a_t, s_{t+1}) - \boldsymbol{\lambda}^\top\boldsymbol{\phi}_t\Big)\right]. \tag{5}$$

Sec. 3.4 specifies r_{d} and the differentiable penalties to avoid sparse events and stabilize gradients across domains.

The formulation highlights the following research issues: (1) *Causal, ego-centric state design* to identify entities influencing control (Sec. 3.3); (2) *Differentiable multi-objective reward shaping* to avoid sparse or discontinuous signals (Sec. 3.4); (3) *Joint uncertainty estimation and use* to guide attention and exploration via entropy gating (Sec. 3.5); and (4) *Policy transfer with causal-uncertainty alignment* for reliable few-shot adaptation (Sec. 3.6).

3.3. Causal State Representation

This subsection instantiates the state $s_t \in \mathcal{S}$ (Sec. 3.1) for the closed-loop task (Sec. 3.2) by exposing entities causally influencing control at time t. Prior fusion encoders compress observations into global tensors, leaving the policy unaware of priorities or confidence, which hampers attribution and safety-driven gradients. The goal is an *ego-centric, compact, differentiable state design* integrating relational cues, ego dynamics, lane geometry, progress, and uncertainty.

Let $\mathcal{E}_t = \{e_i^t\}_{i=1}^M$ denote directed edges from scene entities into the ego node. Each edge carries a compact feature,

$$\mathbf{e}_i^t = \big[\Delta p_i,\ \Delta v_i,\ c_i,\ \kappa_i,\ \sigma_i^2\big], \tag{6}$$

where Δp_i and Δv_i are relative position and velocity in the ego frame, c_i is a semantic code (vehicle, pedestrian, signal), κ_i summarizes local lane geometry (curvature, heading offset), and σ_i^2 is per-entity aleatoric variance. An uncertainty-weighted causal attention assigns relevance,

$$\alpha_i = \text{softmax}_i\left(-\frac{\|\Delta p_i\|_2^2}{\sigma_i^2 + \varepsilon}\right), \quad z_t = \sum_{i=1}^{M}\alpha_i\, W_e\, \mathbf{e}_i^t, \tag{7}$$

with small $\varepsilon > 0$ for numerical stability and W_e a learnable projection. This prioritizes nearer and more confident actors while remaining fully differentiable.

The decision state concatenates the attended interaction embedding with control-relevant scalars,

$$s_t = \big[z_t;\ v_{\text{ego}};\ a_{\text{ego}}^{t-1};\ d_{\text{goal}};\ \phi_{\text{lane}};\ \sigma_{\text{ale}}^2\big]. \tag{8}$$

Here v_{ego} is ego speed, a_{ego}^{-1} the previous action for smoothness, d_{goal} route progress (arc-length to the next target on

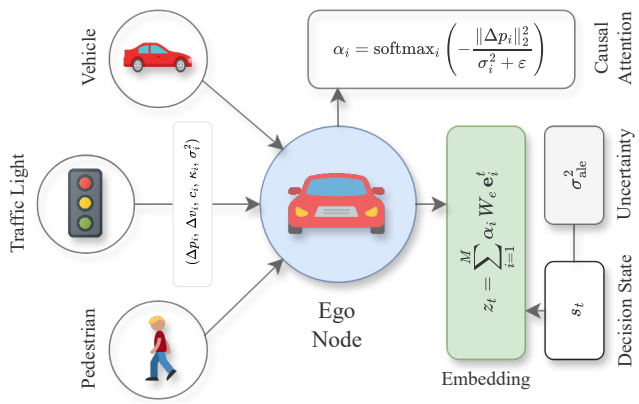

Figure 1. Graph with uncertainty-weighted attention forming s_t.

$\mathcal{R}$), ϕ_{lane} lane-centerline features at p_t (curvature, heading offset consistent with κ_i), and σ_{ale}^2 aggregated aleatoric uncertainty at decision time. Lane-adherence variables (d_L, μ_A) are computed as in Sec. 3.2 and used by reward and penalties, but are not required in s_t at every update.

As shown in Fig. 1, the ego-centric relational graph aggregates incoming edges with uncertainty-weighted attention, emphasizing nearby, confident actors. The attended embedding z_t is fused with ego dynamics, lane geometry, progress, and aleatoric variance to form the control state s_t. This resolves the causal, ego-centric state gap by exposing edge-wise influences and confidence to the policy $\pi_\theta(a|s)$, enabling gradients to capture safety-critical interactions.

3.4. Dense Multi-Objective Shaping

This subsection specifies the dense reward that operationalizes the MDP (Sec. 3.1) and its constraints (Sec. 3.2). The goal is to replace sparse or thresholded events with smooth surrogates that yield stable gradients and transfer across towns, weather, and dynamics. The shaped reward comprises safety, progress, comfort, and uncertainty terms, defined as *differentiable multi-objective reward shaping*.

The per-step reward is a convex combination,

$$r_t \;=\; w_s r_s \;+\; w_p r_p \;+\; w_c r_c \;+\; w_u r_u, \tag{9}$$

with nonnegative weights $w_\cdot$ that satisfy $\sum_\cdot w_\cdot = 1$. It is used as r_{d} in the Lagrangian objective of Eq. (5). All components are continuous, bounded, and differentiable to match clipped actor-critic updates.

Lane adherence, proximity, and signal obedience are encoded with smooth surrogates consistent with Eqs. (2)-(3). Let d_L be lateral deviation with soft barrier $\psi_L(d_L, \mu_A) = \tanh(d_L/\tau_d)$, where corridor membership μ_A shrinks admissible width (Eq. 3). Let $i^\star$ be the nearest actor at distance $\text{dist}(p_t, p_t^{(i^\star)})$, with proximity surrogate $\psi_P = \exp(-\text{dist}/\tau_p)$. Let $\rho_t \in [0, 1]$ denote red-light or stop-line

noncompliance. The safety term is

$$r_s \;=\; 1 \;-\; \kappa_L\,\psi_L(d_L,\mu_A) \;-\; \kappa_P\,\psi_P \;-\; \kappa_R\,\rho_t, \quad (10)$$

with nonnegative coefficients $\kappa_{\{L,P,R\}}$.

Route advancement follows the arc-length coordinate from Sec. 3.2 for progress. Let $\Delta s_t = s^{\text{arc}}_{t+1} - s^{\text{arc}}_t$ denote step progress, and let $\hat{v}_t$ be the projection of ego velocity onto the route tangent. A bounded progress reward is

$$r_p \;=\; \tanh(\Delta s_t/\tau_s) \quad \text{or} \quad r_p \;=\; \tanh(\hat{v}_t/\tau_v), \quad (11)$$

with scales $\tau_s, \tau_v > 0$ ensuring numerical stability.

Longitudinal jerk j_t and steering rate $\dot{\delta}_t$ are penalized quadratically to enforce comfort bounds from Sec. 3.2,

$$r_c \;=\; -\kappa_j\, j_t^2 \;-\; \kappa_\delta\, \dot{\delta}_t^2, \quad (12)$$

where $j_t = \|a^{\text{veh}}_t - a^{\text{veh}}_{t-1}\|_2/\Delta t$ and $\dot{\delta}_t = (a^{\text{str}}_t - a^{\text{str}}_{t-1})/\Delta t$.

Decision-time uncertainty is summarized by $\bar{\sigma} \in [0,1]$ and estimated in Sec. 3.5. A simple, calibrated form encourages confident operation while remaining smooth:

$$r_u \;=\; 1 \;-\; \bar{\sigma}, \quad (13)$$

with optional temperature scaling by validation. This term complements safety shaping without duplicating penalties.

Fig. 2 illustrates the four smooth components and their bounded ranges that stabilize optimization and transfer. The combined reward in Eq. (9) is plugged into Eq. (5) and trained end-to-end with the state in Sec. 3.3.

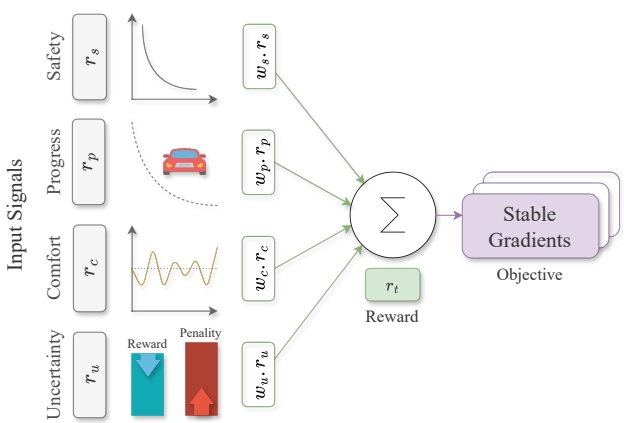

Figure 2. Shaping for safety, progress, comfort, and uncertainty.

3.5. Uncertainty-Gated Exploration

This subsection formalizes decision-time uncertainty for risk-aware, differentiable actor-critic training. Prior designs estimate only one uncertainty source or treat it diagnostically, leaving *policy entropy decoupled* from confidence and degrading safety. The goal is a compact mechanism that jointly estimates aleatoric and epistemic components, propagates them through the ego-relational state (Sec. 3.3), and gates policy entropy in the Lagrangian objective (Eq. 5).

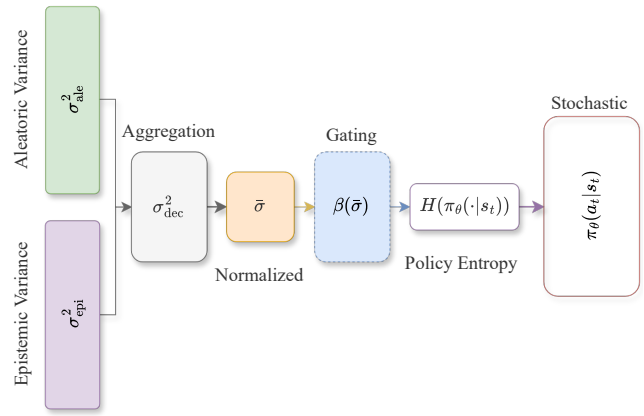

Figure 3. Uncertainty-entropy coupling: σ^2_{ale} from edges, σ^2_{epi} from critic ensemble to normalized uncertainty $\bar{\sigma}$ gates entropy.

Decision-time variance is decomposed as

$$\sigma^2_{\text{dec}} \;=\; \sigma^2_{\text{ale}} \;+\; \sigma^2_{\text{epi}}. \quad (14)$$

Aleatoric variance σ^2_{ale} is estimated per edge in Eq. (6) and injected into the attention of Eq. (7), reducing the weight of distant or noisy actors. Epistemic variance σ^2_{epi} is obtained from an ensemble of critics $\{Q_{\phi_k}\}_{k=1}^K$ via $\sigma^2_{\text{epi}}(s_t,a_t) = \text{Var}_k[Q_{\phi_k}(s_t,a_t)]$, using the same s_t as in Eq. (8). A normalized summary $\bar{\sigma} \in [0,1]$ is produced by temperature-scaled min-max or logistic calibration on a held-out split, yielding a stable confidence signal across domains.

Uncertainty modulates exploration via an entropy term,

$$\mathcal{L}_{\text{ent}} \;=\; -\beta(\bar{\sigma})\, H(\pi_\theta(\cdot|s_t))\,, \quad \beta(\bar{\sigma}) = \beta_0(1-\bar{\sigma}), \quad (15)$$

so low confidence ($\bar{\sigma} \uparrow$) reduces entropy and risk-taking, while high confidence restores exploration. This coupling links the relational state, uncertainty estimates, and policy stochasticity, complementing the uncertainty reward r_u in Eq. (9) without duplicating penalties.

The training objective augments the standard actor-critic loss $\mathcal{L}_{\text{RL}}$ with the entropy gate and Lagrangian shaping:

$$\min_{\theta,\phi} \;\; \mathcal{L}_{\text{RL}}(\theta,\phi;\, r_{\text{d}}) \;+\; \lambda_{\text{ent}}\, \mathcal{L}_{\text{ent}},$$

where r_{d} is given by Eqs. (9)-(13), and penalties enter via Eq. (5). Calibration validates variance heads and the mapping $\bar{\sigma}$ to ensure stable gradients with clipped updates.

Fig. 3 summarizes the flow from per-edge σ^2_{ale} and ensemble σ^2_{epi} to $\bar{\sigma}$ and entropy gating. The next subsection builds on this mechanism to transfer policies by aligning relational attention and uncertainty statistics across domains.

3.6. Causal-Uncertainty Transfer Learning

This subsection addresses policy transfer by aligning control-level semantics and confidence across domains. Adapting only perception features leaves *causal-semantic alignment* and *uncertainty statistics* mismatched at the policy layer, reducing closed-loop generalization. The aim is a compact objective that matches policy distributions, uncertainty-weighted attention, and uncertainty moments, while enabling meta-initialization for few-shot adaptation.

Let π_{θ_s} and π_{θ_t} be source and target policies consuming the same state s_t from Eq. (8). Let $\boldsymbol{\alpha}_d$ denote the attention vector from Eq. (7) in domain $d \in \{s, t\}$, and let u_d summarize uncertainty (moments of $\bar{\sigma}$ and per-edge σ_i^2). Define the policy term $\mathcal{L}_{\mathrm{KL}} = \mathrm{KL}\big(\pi_{\theta_s}(\cdot|s_t) \,\|\, \pi_{\theta_t}(\cdot|s_t)\big)$. The causal-uncertainty transfer loss is

$$\begin{aligned}\mathcal{L}_{\mathrm{trans}} = \mathcal{L}_{\mathrm{KL}} + \lambda_\alpha \,\mathrm{MMD}\big(\boldsymbol{\alpha}_s, \boldsymbol{\alpha}_t\big) \\ + \lambda_u \,\|u_s - u_t\|_2^2,\end{aligned} \tag{16}$$

which aligns action distributions, relational attention, and uncertainty statistics that drive risk-aware behavior.

Initialization applies a MAML-style step to accelerate few-shot adaptation under the Lagrangian objective from Eq. (5) and uncertainty gate in Eq. (15):

$$\theta^\star \;=\; \arg\min_{\theta} \sum_{d \in \mathcal{D}} \mathcal{L}_{\mathrm{RL}}^{(d)}\Big(\theta - \alpha\, \nabla_\theta \mathcal{L}_{\mathrm{RL}}^{(d)}(\theta)\Big)\,, \tag{17}$$

after which a few target updates refine π_{θ_t} while preserving aligned attention and uncertainty profiles.

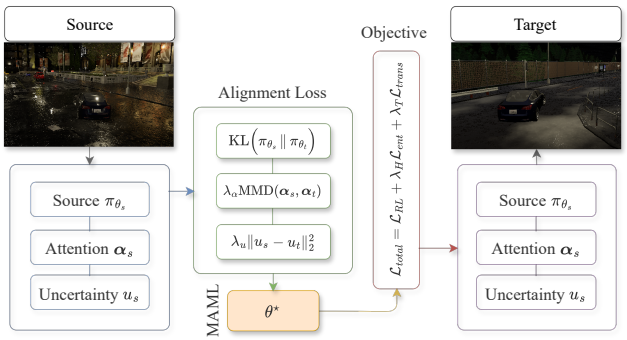

Figure 4. Transfer with KL, MMD, and uncertainty matching.

The full training objective augments actor-critic learning with entropy gating and transfer regularization:

$$\min_{\theta,\phi}\; \mathcal{L}_{\mathrm{RL}}(\theta, \phi; r_{\mathrm{d}}) + \lambda_{\mathrm{ent}}\, \mathcal{L}_{\mathrm{ent}} + \lambda_T\, \mathcal{L}_{\mathrm{trans}}, \tag{18}$$

where r_{d} is defined in Sec. 3.4 and $\mathcal{L}_{\mathrm{ent}}$ in Eq. (15). This formulation ensures that control-relevant semantics and confidence calibration governing exploration remain consistent across towns, traffic, and weather. Fig. 4 summarizes the causal-uncertainty alignment from source to target, coupled with meta-initialization for rapid adaptation.

4. Closed-Loop Validation and Assessment

This section validates the proposed framework in a closed loop and links the quantitative outcomes to the design choices in Secs. 3.3–3.6. Evaluation focuses on safety, stability, and generalization across town layouts and adverse weather, and compares against strong E2E baselines ST-P3 [10], ThinkTwice [14], TransFuser [5], and RaSc [8]. Results are reported first for the influence of the ego-relational state, then for dense reward shaping, uncertainty-gated exploration, and finally for policy transfer.

4.1. Simulation Protocol and Metrics

Experiments are conducted in CARLA 0.9.15 with synchronous stepping at 20 Hz and a Tesla Model 3 ego vehicle. The agent controls continuous throttle, brake, and steering as in Sec. 3.1, and uses Soft Actor-Critic (SAC) with the entropy-gated objective in Sec. 3.5 and the Lagrangian shaping in Eq. (5). Training happens in Town10HD under a deliberately adverse regime. Weather is set to heavy rain and dense fog at night (cloudiness 90%, precipitation 90%, fog density 40%, sun altitude $-25°$). NPC traffic density is drawn from a moderate-to-high range for each episode.

Hyperparameters follow standard SAC practice. Replay buffers store 2×10^5 transitions. Batches of size 512 are sampled with discount factor $\gamma = 0.99$ and target update coefficient $\tau = 5 \times 10^{-3}$. The actor and critics use Adam with a learning rate of 3×10^{-4}. The entropy weight is modulated by normalized uncertainty $\bar{\sigma}$ via $\beta(\bar{\sigma}) = \beta_0(1 - \bar{\sigma})$, with $\beta_0 \in \{0.5, 1.0\}$ selected on a validation split. Training runs for 5×10^5 steps with evaluation every 10k steps.

Closed-loop statistics follow the definitions in Sec. 3.2. Collision, off-road, and timeout rates are measured per kilometer as $\mathrm{q/km} = N_{\mathrm{q}}/D_{\mathrm{km}}$, where D_{km} is the total distance traveled in kilometers and N_{q} is the count of infractions of type $q \in \{\mathrm{coll}, \mathrm{off}, \mathrm{to}\}$. Route completion is computed as $\mathrm{RC}(\%) = 100 \cdot (\text{traversed arc-length}/\text{route arc-length})$. The success rate is defined as $\mathrm{SR} = N_{\mathrm{succ}}/N_{\mathrm{ep}}$.

A multiplicative Driving Score (DS) measures the trade-off between progress and infractions. It is defined as $\mathrm{DS} = 100 \cdot \mathrm{RC} \cdot \prod_{q \in \{\mathrm{coll},\mathrm{off},\mathrm{to}\}} \Big(1 - \min\Big(1, \frac{\mathrm{q/km}}{\kappa_q}\Big)\Big)$, with saturation constants $\kappa_{\mathrm{coll}} = \kappa_{\mathrm{off}} = \kappa_{\mathrm{to}} = 0.02$. The multiplicative penalty term is also reported as the Infraction Score (IS). Driving stability is evaluated using Cross-Track Error (CTE) and heading error as in Eq. (2), which directly reflect the lane-related variables shaped in Sec. 3.4.

4.2. Validation of Ego-Relational State

The first set of experiments evaluates the causal ego-relational state in Sec. 3.3. The comparison uses baselines that differ in state encoding: ST-P3 [10] and ThinkTwice [14] rely on spatial-temporal tensors, TransFuser [5] fuses BEV and camera features, while RaSc [8] augments

imitation with risk awareness but still employs global maps. The proposed state replaces global tensors with an ego-node relational graph whose edges encode relative kinematics, lane geometry, semantics, and per-entity aleatoric variance, aggregated via uncertainty-weighted attention in Eq. (7) into the compact control state s_t in Eq. (8).

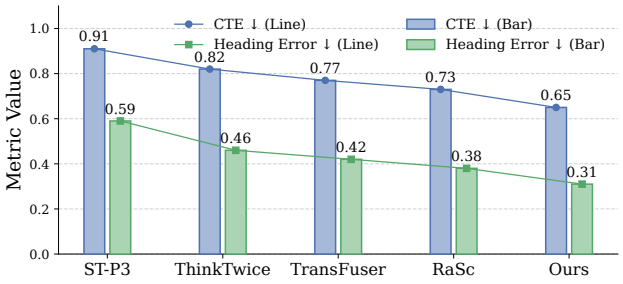

Figure 5. Ego-relational state ablation on Town10HD, CTE, and heading error reduced with uncertainty-weighted relational state.

Fig. 5 reports CTE and heading error on Town10HD. The causal state reduces lateral deviation compared to tensor and BEV-based encoders. Average CTE drops to 0.65, improving 28.6% over ST-P3, 20.7% over ThinkTwice, and 15.6%/11.0% over TransFuser and RaSc. Heading error falls to 0.31, a 47.5% reduction from ST-P3, with 32.6%, 26.2%, and 18.4% gains over ThinkTwice, TransFuser, and RaSc. These improvements reflect the design of Eq. (8), where lane geometry and ego dynamics are fused with uncertainty-weighted interaction features.

Fig. 6 shows route completion via off-road percentage and goal rate. Off-road drops from 10.8% (ST-P3) to 4.1% (62% reduction), with 55%, 46%, and 37% gains over ThinkTwice, TransFuser, and RaSc. Goal completion rises to 79.5%, improving 35.4% over ST-P3, 23.6% over ThinkTwice, and 14.4%/10.3% over TransFuser and RaSc.

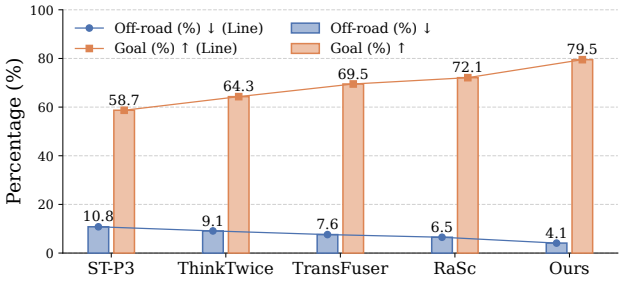

Figure 6. Route completion metrics. Off-road (%) decreases and goal rate increases when the policy uses the causal state.

These trends show that the uncertainty-weighted relational state improves stability and goal attainment, reducing off-road and boosting success compared to tensor baselines.

4.3. Impact of Dense Reward Shaping

The second set of experiments studies the dense multi-objective shaping in Sec. 3.4. The baseline methods use progress-dominated or event-driven rewards, with discontinuous penalties for collisions and lane departures. The proposed formulation replaces such thresholds with differentiable surrogates for safety, progress, comfort, and uncertainty, as summarized in Eqs. (9)–(13), so that gradients remain informative even under rare safety events.

Fig. 7 compares average episode rewards on Town10HD. ST-P3 achieves 156.4, ThinkTwice 164.1, TransFuser 174.2, and RaSc 182.9. The uncertainty-aware DRL framework with dense shaping reaches 265.3, improving returns by 69.6% over ST-P3, 61.7% over ThinkTwice, 52.3% over TransFuser, and 45.1% over RaSc. Learning curves show smoother growth and fewer oscillations, indicating that differentiable penalties in Eq. (10) and comfort regularization in Eq. (12) reduce sensitivity to rare large-magnitude events.

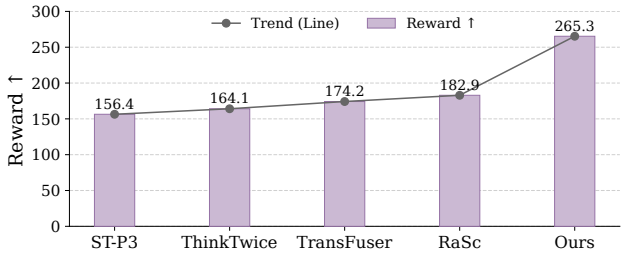

Figure 7. Reward comparison where differentiable multi-objective shaping yields higher, more stable returns than baselines.

Reward gains improve safety and comfort in a closed loop. Continuous surrogates in r_s and r_p let the critic adapt even in near-miss cases. Jerk and steering-rate penalties (Eq. 12) reduce oscillations, lowering CTE and heading error. The uncertainty term r_u (Eq. 13) promotes well-observed states, making rewards robust to sensor noise.

4.4. Uncertainty-Gated Exploration Ablation

The third set of experiments isolates the impact of joint aleatoric-epistemic uncertainty estimation and entropy gating in Sec. 3.5. Baselines ignore uncertainty or treat it diagnostically without feedback to attention or exploration. The proposed design propagates aleatoric variance through attention weights, estimates epistemic variance via a critic ensemble, and compresses both into the normalized summary $\bar{\sigma}$ for the entropy gate in Eq. (15).

Fig. 8 reports exploration variance, collision rate, and stability. ST-P3 is normalized to 1.00, while ThinkTwice, TransFuser, and RaSc reach 0.89, 0.80, and 0.72. The uncertainty-gated policy reduces variance to 0.62, a 38% drop vs. ST-P3 and 30.3%, 22.5%, 13.9% vs. ThinkTwice, TransFuser, and RaSc. This stems from the entropy penalty

in Eq. (15), which lowers randomness when $\bar{\sigma}$ indicates low confidence, especially under fog or occlusion.

Collision rates drop from 0.011 (ST-P3), 0.009 (ThinkTwice), 0.008 (TransFuser), and 0.007 (RaSc) to 0.006 with uncertainty-gated exploration, 45.5%-14.3% fewer than baselines. Stability rises to 0.91, exceeding ST-P3 (0.72), ThinkTwice (0.77), TransFuser (0.82), and RaSc (0.86).

Results validate the coupling of uncertainty with exploration: variance and critic regulate attention and entropy, making the agent cautious under ambiguity and exploratory when confident. Collision and stability gains show improved safety-exploration balance over fixed schedules.

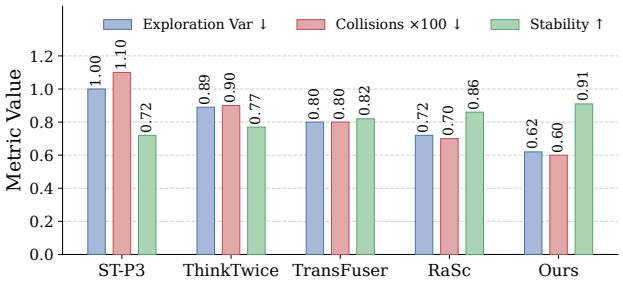

Figure 8. Uncertainty-aware exploration and safety metrics.

4.5. Impact of Causal-Uncertainty Transfer

The final experiments study policy transfer from Town10HD (source) to Town02 (target) using the causal-uncertainty objective in Sec. 3.6. The source involves adverse weather with dense traffic, while the target adds distinct topology, intersections, and mixed weather. Three variants are compared: (i) Policy Learning trains a SAC agent on the target without transfer regularization; (ii) Source Domain evaluates the Town10HD policy without adaptation; (iii) Target Domain applies causal-uncertainty transfer, aligning distributions, attention, and uncertainty via Eq. (16) with MAML initialization in Eq. (17).

Tab. 1 reports averages over 20 episodes. Driving Score rises from 188.6 (policy learning) to 205.7 (source domain) and 214.3 (transfer in target domain). Route completion improves from 75.2% to 82.6% and 84.1%. Collisions/km fall from 0.007 to 0.006 and 0.005. Off-road/timeout drop from 0.005/0.003 to 0.004/0.002 and 0.003/0.001. Infraction Score increases from 0.88 to 0.92 and 0.94, showing transfer achieves higher progress with fewer infractions.

To connect aggregated metrics with dynamics, per-episode statistics are extracted from logs. On the source domain, 1,693 episodes yield reward 265.3, mean CTE 0.65, and heading error 0.26. On the target domain, 100 adaptation episodes achieve a reward 558.9, with CTE 0.73 and reduced heading error 0.18. Despite differing geometry and traffic, causal-semantic alignment (Eq. 16) and meta-initialization (Eq. 17) stabilize attention and uncertainty.

Table 1. Effectiveness of causal-uncertainty transfer from Town10HD to Town02. Values are averaged over 20 episodes.

Variant	DS	RC (%)	Coll./km	OR/km	TO/km	IS
Policy	188.6	75.2	0.007	0.005	0.003	0.88
Source	205.7	82.6	0.006	0.004	0.002	0.92
Target	**214.3**	**84.1**	**0.005**	**0.003**	**0.001**	**0.94**

Transfer results show that aligning action distributions, attention, and uncertainty across domains improves success and safety over target-only training. Unlike perception adapters [13, 20], the objective regularizes the decision layer where causality and confidence guide control. The policy retains ego-relational and uncertainty-aware benefits while narrowing the generalization gap across towns.

Overall, evaluations show the ego-relational state improves stability, the multi-objective reward reduces collisions, uncertainty-gated exploration ties behavior to confidence, and causal-uncertainty transfer boosts scores across towns. Opportunities include reliance on simulation-to-simulation transfer, overhead from critic ensembles, tuning of reward and entropy parameters, and attributional rather than interventional causal modeling. Extending to real vehicles and multi-agent settings is a natural next step.

5. Conclusion

This work presented a reliability-centered Deep RL framework that couples a causal ego-relational state, differentiable multi-objective shaping, and joint aleatoric-epistemic estimation with entropy-gated exploration and causal-uncertainty transfer. Closed-loop evaluation in CARLA across adverse weather, dense traffic, and cross-town transfer demonstrated consistent gains in stability, driving score, and infraction rates over strong end-to-end baselines, indicating that control-level reasoning benefits from uncertainty-weighted interaction structure rather than purely perception-centric encoders. Dense rewards aligned with lane-keeping, progress, comfort, and confidence yielded smoother optimization and more predictable behavior, while uncertainty-gated entropy reduced collisions without sacrificing route completion, supporting the design of coupling confidence and exploration. Causal-uncertainty alignment across source and target towns improved policy transfer with few-shot adaptation, suggesting that matching action distributions, attention patterns, and uncertainty statistics is an effective strategy for mitigating distribution shift at the decision layer. The overall study highlights that safety-aware policy transfer in autonomous driving can be systematically addressed by jointly shaping state, reward, and uncertainty, and it motivates future extensions toward real-vehicle deployment.

Acknowledgements

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